A quotient that excludes the four-point suffix

Human Compression project: L044 continuation

Lemma 1 (A fixed point preserves the quotient intersection property)**.** *Let $H = \langle h_0, \dots, h_k \rangle$ be a string C-group, $k \geq 2$, and let $\pi : H \longrightarrow \overline{H}$ be a homomorphism to a permutation group on X. Suppose*

$$N = \ker \pi \leq \langle h_{k-1}, h_k \rangle, \qquad \pi \text{ is injective on } M = \langle h_0, \dots, h_{k-1} \rangle.$$

Suppose some $b \in X$ is fixed by $\pi(M)$ and moved by $\pi(h_k)$. Then the ordered projected generators form a string C-group of rank $k+1$.

Proof. Every projected mark is a nonidentity involution: this follows from injectivity on M for the first k marks and from the motion of b for the last. String commutation is preserved. Write $H_I = \langle h_i : i \in I \rangle$ and $T = \{k-1, k\}$. We prove the intersection property for arbitrary index sets I, J.

If $T \subseteq I$, then $N \leq H_I$. For $x \in H_I$ and $y \in H_J$ with $\pi(x) = \pi(y)$, the element yx^{-1} lies in $N \leq H_I$, hence $y \in H_I \cap H_J$. The original IP gives $\pi(H_I) \cap \pi(H_J) = \pi(H_{I \cap J})$. The same proof applies when $T \subseteq J$. This step uses containment of the entire kernel, not merely a nontrivial intersection with it.

If $T \not\subseteq I \cup J$, the original IP gives

$$N \cap H_{I \cup J} \leq H_T \cap H_{I \cup J} = H_{T \cap (I \cup J)}.$$

The last subgroup is trivial or generated by one of the two last marks. Each such mark has nonidentity image, so its intersection with N is trivial. Thus π is injective on $H_{I \cup J}$, and its intersection property follows directly from that of H.

In the remaining case $T \subseteq I \cup J$ but neither set contains T. Interchanging I, J if necessary, one has $k \in I$, $k-1 \notin I$, $k-1 \in J$, $k \notin J$. All marks in $I \setminus \{k\}$ commute with h_k, so any $x \in H_I$ has the form $x = a h_k^{\epsilon}$, where $a \in H_{I \setminus \{k\}} \leq M$ and $\epsilon \in \{0, 1\}$. If $\pi(x) = \pi(y)$ for $y \in H_J \leq M$, the image $\pi(x)$ fixes b. Since $\pi(a)$ fixes b and $\pi(h_k)$ moves it, one must have $\epsilon = 0$. Both x, y now lie in M, where π is injective. Thus $x = y \in H_I \cap H_J$, and the required equality follows again.

This proves IP for all subsets. In particular no projected mark belongs to the subgroup generated by the other projected marks: applying IP to those complementary index sets would make its nonidentity cyclic subgroup trivial. The stated rank is therefore exactly $k+1$. □

Theorem 2 (The decorated four-point suffix is impossible)**.** *Retain the complete critical-head hypotheses and pure-prefix theorem, in the contrary range $n = 2r$. Assume explicitly, as the strong-induction hypothesis, that for every $12 \leq m < n$ the natural alternating group A_m has no string C-marking of rank greater than $\lfloor (m-1)/2 \rfloor$.*

Suppose $g \geq 5$, $r = g+3$, and $\Omega = C \dot{\cup} A$, where $|A| = 4$. Let $P_i = G_{\leq i}$ and suppose P_{g-1} fixes $A \cup \{b\}$ pointwise, $b \in C$, while P_g is transitive on C and contains its pure natural alternating group. Suppose the last three actual marks are

$$\begin{aligned} q &= \rho_{g+2} = (a_0\ a_1)(a_0'\ a_1'), \qquad A = \{a_0, a_0', a_1, a_1'\}, \\ s &= \rho_{g+1} = (b\ a_0)(a_1\ a_1'), \\ u &= \rho_g = (b\ c)(a_0\ a_0')(a_1\ a_1')\delta, \qquad c \in C \setminus \{b\}, \end{aligned}$$

where δ is supported on $C \setminus \{b, c\}$. Then no such ambient string C-group exists.

Proof. Put $e = \rho_{g-1}$ and $v = (a_0\ a_0')(a_1\ a_1')$. Every earlier mark ρ_i, $i \le g-2$, fixes c: it commutes with u and fixes b. The generator e must move c. Otherwise all earlier marks would fix $A \cup \{b, c\}$ pointwise and the last three marks would preserve it, contrary to ambient transitivity. Write $y = c^e \in C \setminus \{b, c\}$.

The complete acquired seven-point alternating-subgroup lemma applies to the reversed four marks q, s, u, e. For exactness, its labels $1, 2, 3, 4, 5, 6$ correspond respectively to $a_1, a_0, a_1', a_0', b, c$, so the three literal permutations in that lemma become exactly q, s, u above. Its last involution has the form $e = (c\ y)\epsilon$ and fixes $A \cup \{b\}$, as required. The lemma permits arbitrary disjoint external factors δ, ϵ; no natural alternating-tail assumption from the separate theorem in that packet is used. Consequently

$$\mathrm{Alt}(A \cup \{b, c, y\}) \le G_{\{g-1,g,g+1,g+2\}}. \tag{1}$$

In particular the pure even permutation v belongs to this actual suffix.

On the other hand, the exterior image of P_g on A is exactly $\langle v \rangle$: every earlier mark fixes A, and $u|_A = v$. This exterior image is even. Since every ambient permutation is even, the image of P_g on C is contained in $\mathrm{Alt}(C)$. Its pure alternating kernel is already the whole $\mathrm{Alt}(C)$, by hypothesis. Multiplying u by the inverse of its pure C-restriction therefore gives $v \in P_g$, and in fact

$$P_g = \mathrm{Alt}(C) \times \langle v \rangle. \tag{2}$$

Actual ambient IP applied to (1) now gives

$$v \in P_g \cap G_{\ge g-1} = \langle e, u \rangle. \tag{3}$$

Project (2) to its action on C. The entire kernel is $N = \langle v \rangle$, and (3) puts that kernel in the last two marks of P_g. Projection is injective on P_{g-1}, since that group fixes A pointwise, whereas v does not. Finally P_{g-1} fixes b and u moves b to c. Lemma 1 therefore proves that the actual image $\mathrm{Alt}(C)$ has a string C-marking of rank $g+1$. Every projected mark is nonidentity, and independence is included in that lemma; no rank is inferred merely by counting the original marks before projection.

Here $n = 2r = 2g+6$, so $|C| = n-4 = 2g+2$. The assumption $g \ge 5$ gives $12 \le |C| < n$. The explicit smaller-degree induction hypothesis applies and requires rank at most $\lfloor (|C|-1)/2 \rfloor = g$. This contradicts rank $g+1$ and excludes the stated configuration. □

Attribution and remaining scope. The seven-point alternating subgroup is an acquired proof, used with its exact literal substitution; its separate natural-tail theorem is not invoked. The pure-prefix argument and its disclosed root parity-fibre contribution remain paid. The new quotient lemma uses a fixed point to verify every projected intersection, closing the decorated case left by the small-suffix reduction. Root requested explicit checks of the induction hypothesis, full kernel containment and projected rank during review preparation; those checks are included above. The literal suffix is a theorem hypothesis, so this packet does not assume an unreviewed small-suffix derivation. Other prefix-orbit configurations, endpoint cuts, opposite heads, terminal singleton and the complete L044 recovery remain separate. No new computation, formal theorem, whole-original or question credit is claimed.